

Updated California Aftershock Parameters

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ABSTRACT

Reasenberg and Jones (1989) introduced a statistical model for aftershock rate following a mainshock along with estimates of “generic” California parameter values based on past aftershock sequences. The Reasenberg and Jones (1989) model has been used for decades to issue aftershock forecasts following $M \geq 5$ mainshocks in California. Here, we update the “generic” parameters for California through a fit to the aftershock sequences of $M \geq 5$ mainshocks occurring since 1980. We find aftershock productivity values that are lower on average than the generic productivity reported by Reasenberg and Jones (1989), likely because low-productivity sequences were omitted from their analysis and possibly because of a trade-off between productivity and b -value. We confirm the observation of Llenos and Michael (2017) that southern California sequences are more productive on average than northern California sequences. The Mendocino area is much less productive; the hydrothermal areas in Long Valley, Coso, and the Salton Sea, in contrast, are much more productive. We also quantify the variability of the Reasenberg and Jones (1989) productivity parameter a between sequences with a normal distribution. This distribution of a -values can be used to compute aftershock forecasts that include epistemic uncertainty and can be used as the prior for Bayesian updating of the a -value as a sequence progresses.

INTRODUCTION

Reasenberg and Jones (1989) introduced a statistical method to forecast aftershocks by combining the modified Omori decay of aftershocks (Utsu, 1957), the Gutenberg–Richter magnitude–frequency distribution (Gutenberg and Richter, 1944), and the Utsu aftershock productivity (Utsu, 1972). In their model, the rate $\lambda(t, M)$ of magnitude $\geq M$ aftershocks at time t after a mainshock of magnitude M_m is given as

$$\lambda(t, M) = 10^{a+b(M_m-M)}(t+c)^{-p}, \quad (1)$$

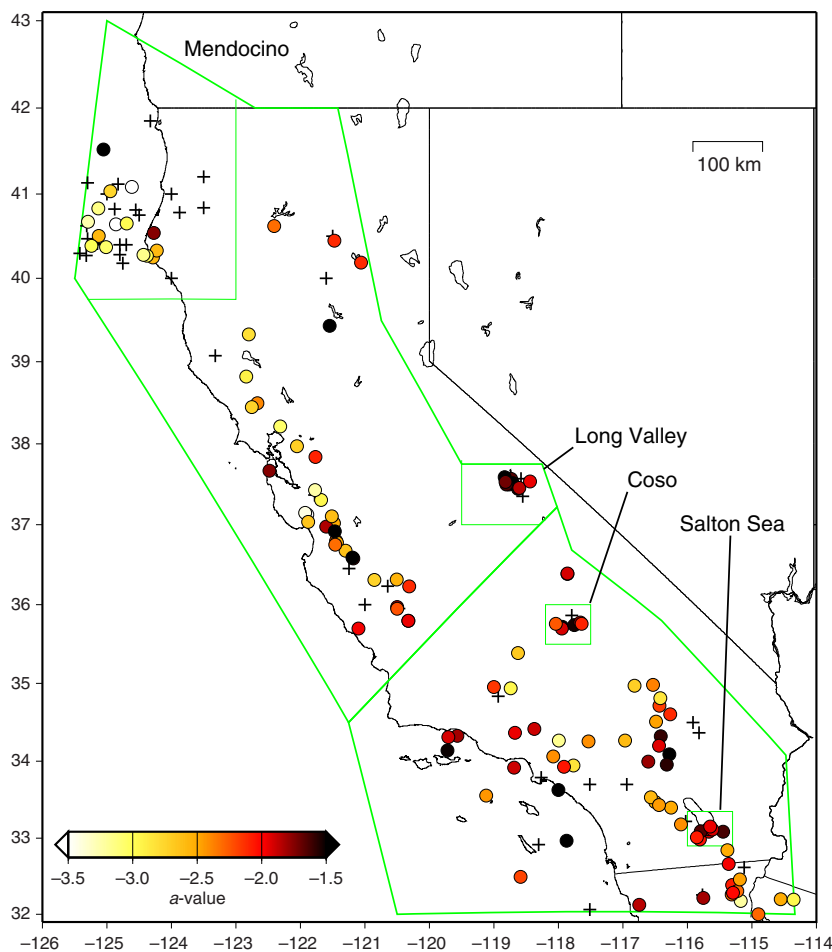
in which b is the b -value of the magnitude–frequency distribution, a is a parameter representing the aftershock productivity, p is a parameter representing the temporal decay of the sequence, and c is the time scale of the early part of the sequence before the beginning of the temporal decay. A similar equation can also be derived by constraining the productivity

using Båth’s law for the average size of the largest aftershock (Shcherbakov *et al.*, 2004).

Reasenberg and Jones (1989) fit equation (1) for a number of aftershock sequences in California, finding median parameter values of $a = -1.67$, $b = 0.91$, $p = 1.08$, and $c = 0.05$ days, which they term the “generic” California parameters. They proposed that aftershock forecasts can be made for California mainshocks using these generic parameters. After enough aftershocks have occurred, “sequence-specific” parameter values can be determined for the sequence in progress, and the generic- and sequence-specific estimates for each parameter can be combined using Bayesian estimation. The Reasenberg and Jones (1989) method has been used for decades by the California Integrated Seismic Network (CISN) to issue aftershock forecasts following $M \geq 5$ mainshocks in California.

The U.S. Geological Survey (USGS) now issues similar aftershock forecasts for the entire United States. To make this possible, Page *et al.* (2016) extended the Reasenberg and Jones (1989) method to apply globally by grouping tectonically similar areas together and determining generic parameter values for each group. They determined values of a and p for each global tectonic region from a joint inversion of all aftershock sequence in the region, fixing $b = 1.0$ and $c = 0.018$ days. Page *et al.* (2016) also determined a -values for individual sequences, fixing the p -value to the value found from the joint inversion because fitting of both a and p for a single sequence can be unstable (e.g., Llenos and Michael, 2017). Page *et al.* (2016) then found the underlying distribution of a -values for each tectonic region using an inversion method to account for discretization of the observed a -values due to sequences with no or only very few aftershocks. This distribution can then be used to generate forecasts with confidence bounds that reflect both the epistemic uncertainty (the distribution of a -values) and the aleatory variability (modeled as a Poisson process). This distribution can also be used as the prior for Bayesian updating of the a -value distribution as the sequence progresses.

We revisit and update the “generic” parameters for California. Following the work of Page *et al.* (2016), we also determine the distribution of a -values to characterize the epistemic uncertainty and for use as a prior in Bayesian updating. We now have almost three decades more data than available to Reasenberg and Jones (1989), much of it also of higher quality. Additionally, their California median productivity value of $a = -1.67$ is no longer used in practice. The computer code used in CISN operations uses instead the mean value of



▲ **Figure 1.** Map of California, with polygons showing the Southern California Seismic Network (SCSN) and the Northern California Seismic System (NCSS) authoritative regions, as well as the Mendocino region and the three hydrothermal areas: Long Valley, Coso, and the Salton Sea. Circles show a , the productivity parameter of the [Reasenber and Jones \(1989\)](#) model, found for individual aftershock sequences of $M \geq 5$ mainshocks from 1933 to 2017. Plus symbols indicate a mainshock with no aftershocks in the first 10 days with magnitude above the magnitude of completeness, so no a -value could be determined. The color version of this figure is available only in the electronic edition.

$a = -1.76$, and many studies opt to use a lower value of $a = -1.85$ (e.g., [Michael, 2012](#)). The original a -value in [Reasenber and Jones \(1989\)](#) was likely too high because they omitted sequences that did not have enough aftershocks to fit both the a -value and p -value, effectively omitting the sequences with the lowest a -values.

DATA AND METHOD

We use the earthquake catalog of the CISN, from 1932 to July 2017. Earthquake locations and magnitudes contributed by the Southern California Seismic Network (SCSN) and the Northern California Seismic System (NCSS) are used within their respective authoritative regions. We determine parameters separately for the SCSN and NCSS regions because [Llenos and Michael](#)

(2017) found a higher productivity for strike-slip mainshocks in southern California compared with northern California. We also determine separate parameters for the region of the Mendocino triple junction and for hydrothermal regions in Long Valley, Coso, and the Salton Sea (Fig. 1), excluding these areas from the SCSN and NCSS regions. The Geysers remains included with the NCSS region because its productivity appears to be lower than the other hydrothermal regions.

We estimate the catalog magnitude of completeness M_{cat} as a function of year and region. We find the magnitude of completeness for the later parts of individual large aftershock sequences using the maximum curvature method with a correction of 0.2 magnitude unit ([Woessner and Wiemer, 2005](#)), verifying each result by inspection and adjusting by hand if necessary. We then determine a minimal number of time intervals with monotonically decreasing M_{cat} that fits most of the sequences (Table 1). The magnitude of completeness estimated by [Felzer \(2008\)](#) is used when estimates from sequences are not available. We also assume that the aftershock sequence of the 2010 M_w 7.2 El Mayor earthquake is incomplete at all magnitudes because it falls partially outside the CISN authoritative region. We assume that the magnitude of completeness M_c following a mainshock is elevated above M_{cat} , in accordance with the time-dependent magnitude of completeness found for California ([Helmstetter et al., 2006](#); [Hainzl, 2016](#))

$$M_c(t, M_m, M_{cat}) = \max(M_m - 4.5 - 0.75 \log_{10}(t), M_{cat}), \quad (2)$$

in which t is the time since the magnitude M_m mainshock.

We define mainshocks as all earthquakes $M \geq 5$, excluding events that occur in the three years following a larger event and within five times the [Wells and Coppersmith \(1994\)](#) rupture length of that larger event. There were 170 mainshocks since 1932. Aftershocks are defined as all events within 10 days after a mainshock and three times the [Wells and Coppersmith \(1994\)](#) rupture length of the mainshock. The smallest radius of an aftershock zone is 10 km for an M 5.0 mainshock, so selecting a smaller radius might exclude aftershocks because of location errors, especially for older events. We nevertheless also try using a smaller radius of one or two times the [Wells and Coppersmith \(1994\)](#) rupture length and find that the results are not substantially changed. The relatively short time window of 10 days reduces contamination of the aftershocks with background seismicity, so using the larger aftershock region should ensure that most aftershocks are captured while including only a negligible number of background events.

Table 1 Magnitude of Completeness for Various Regions and Time Periods		
Region	Years	M_{cat}
Long Valley	1932–1977	4.5
	1978–1979	3.5
	1980–1983	3.3
	1984–present	2.3
Mendocino	1932–1941	5.6
	1942–1951	5.2
	1952–1956	5.1
	1957–1974	4.7
	1975–1990	4.0
	1991–present	2.6
Remaining CISN authoritative region	1932–1940	4.0
	1941–1959	3.5
	1960–1975	3.3
	1976–present	2.3
CISN, California Integrated Seismic Network.		

We invert for the best set of aftershock parameters for each region, jointly fitting all mainshocks occurring from 1980 to 2017. Older events are excluded because many have no or few recorded aftershocks and high M_{cat} , which puts little constraint on the underlying parameter values. From 1980 to 2017, there were 23 mainshocks in the SCSN region, 16 in the NCSS region, 16 in the Mendocino region, and 11 in the hydrothermal areas. For each region, we find the parameter values that maximize the likelihood L

$$L = \prod_{j=1}^K \prod_{i=1}^N \lambda(t_i, M_c(t_i, M_{mj}, M_{\text{cat}})) \times \exp \left\{ - \int_{t_0}^{t_1} \lambda(t, M_c(t, M_{mj}, M_{\text{cat}})) dt \right\}. \quad (3)$$

The likelihood is defined as the probability of observing the aftershocks given the model (equation 1) and the set of parameter values (Ogata, 1983). The first term gives the probability of an event occurring at all of the N aftershock times, which is the product of the probability of occurrence at each aftershock time. The second term gives the probability of no events occurring during the interevent periods, which is equivalent to the entire time period given the infinitesimal duration of an earthquake in a point process model. The probabilities for each of the K sequences are then multiplied together to determine the overall probability. We use the log likelihood (LL):

$$\text{LL} = \sum_{j=1}^K \left[\sum_{i=1}^N \log \lambda(t_i, M_c(t_i, M_{mj}, M_{\text{cat}})) - \int_{t_0}^{t_1} \lambda(t, M_c(t, M_{mj}, M_{\text{cat}})) dt \right]. \quad (4)$$

The times t_0 and t_1 indicate the time frame of the sequences, here 0 to 10 days. The maximum LL is found through a grid search over the three parameters a , p , and c , fixing $b = 1.0$. Page *et al.* (2016) include a constraint that the rate $\lambda(t, M)$ cannot be lower than the rate assuming $M_c = M_{\text{cat}}$. We find that the rate found assuming $M_c = M_{\text{cat}}$ is always similar or lower, so no inversion constraint is needed. We test for consistency in productivity over time by inverting the events from 1932 to 1979 for the a -value only and fixing the other parameters to the values found for events from 1980 to 2017.

Next, we determine a -values for the 170 individual sequences from 1932 to 2017 by maximizing LL as a function of a , and keeping the p -value and c -value fixed to the values found from the joint inversion for the region. We fit only the a -value for individual sequences because Llenos and Michael (2017) demonstrated that fitting multiple parameters for a single sequence can be unstable. Additionally, with p and c fixed, we are able to estimate a -values for sequences with very few aftershocks. The 62 sequences with no aftershocks $\geq M_c(t, M_m, M_{\text{cat}})$ within the first 10 days are retained, although the a -value cannot be estimated. Many of the smaller mainshocks before 1980 fall into this category because of higher M_{cat} . We estimate the a -value uncertainty with respect to the location and magnitude uncertainty. To do this, we recalculate the a -values for each sequence 1000 times, with the location of each potential aftershock randomly perturbed by the assumed location uncertainty (± 10 km before 1960, ± 5 km from 1960 to 1980, and ± 2 km after 1980) and the magnitude randomly perturbed by the assumed magnitude uncertainty (± 0.222 before 1973, and ± 0.111 after 1973) from Felzer and Cao (2008). We define the uncertainty as the middle 68% of recomputed a -values. We find that the root mean square a -value uncertainties over all events are 0.43 before 1980 and 0.05 after 1980.

Finally, we use the distribution of observed a -values for individual sequences to estimate the underlying distribution of true a -values for each region. The underlying a -value distribution may be obscured because of sparse sampling by the limited number of mainshocks and because low a -values may be poorly constrained due to sequences with no or only very few aftershocks. We follow the inversion method introduced by Page *et al.* (2016) to transform the observed a -values to their probable true underlying distribution. The first step of the inversion is to determine the matrix $\mathbf{A}$ that maps the discretized true value of a to the discretized probability distribution for the observed value of a . We find the $\mathbf{A}$ matrix using 10,000 synthetic aftershock catalogs for each “true” value of a . We randomly generate each synthetic aftershock catalog assuming the Reasenber and Jones (1989) model with the given a -value and a mainshock with a given magnitude M_m . The time-dependent magnitude of completeness $M_c(t, M_m, M_{\text{cat}})$ is applied, with pairs of M_m and M_{cat} values randomly chosen from the real catalog. We then determine the “observed” a -value that maximizes LL for the synthetic catalog. The normalized distribution of the 10,000 “observed” a -values for the “true” a -value forms a row of the $\mathbf{A}$ matrix.

The underlying a -value distribution $\mathbf{x}$ can then be found from the observed distribution $\mathbf{d}$, by solving $\mathbf{Ax} = \mathbf{d}$. The $\mathbf{x}$ and $\mathbf{d}$ vectors represent the a -values as discretized probability distributions, with the same discretization as used for the $\mathbf{A}$ matrix. A simple flattening constraint is also applied to minimize the difference between adjoining a -values in the vector $\mathbf{x}$ using a damping matrix $\mathbf{D}$

$$\mathbf{D} = \begin{pmatrix} 1 & -1 & 0 & \dots \\ 0 & 1 & -1 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}, \quad (5)$$

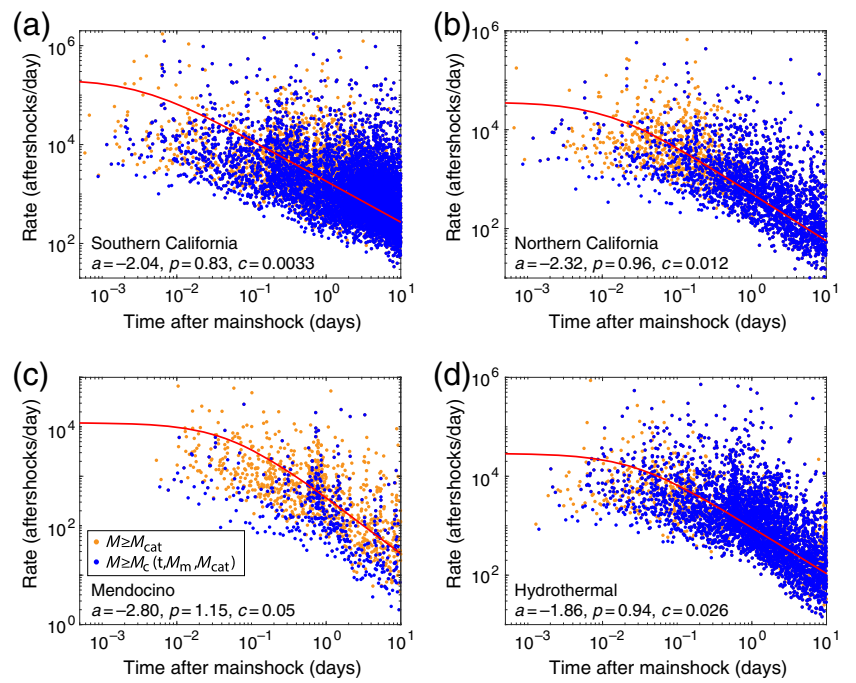
and then solving the equation

$$\begin{pmatrix} \mathbf{A} \\ e\mathbf{D} \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{d} \\ 0 \end{pmatrix}, \quad (6)$$

for $\mathbf{x}$ using nonnegative least squares. The damping parameter e is selected to best jointly minimize the misfit and model length. We use all mainshocks occurring from 1980 to 2017 because the older events include many sequences with high M_{cat} and no recorded aftershocks, and those with aftershocks exhibit a larger uncertainty in a -value. Page *et al.* (2016) apply the inversion to subsets of their data with different ranges of mainshock magnitude, but we have too few mainshocks in California to consider subsets, and therefore invert all magnitude ranges together.

RESULTS

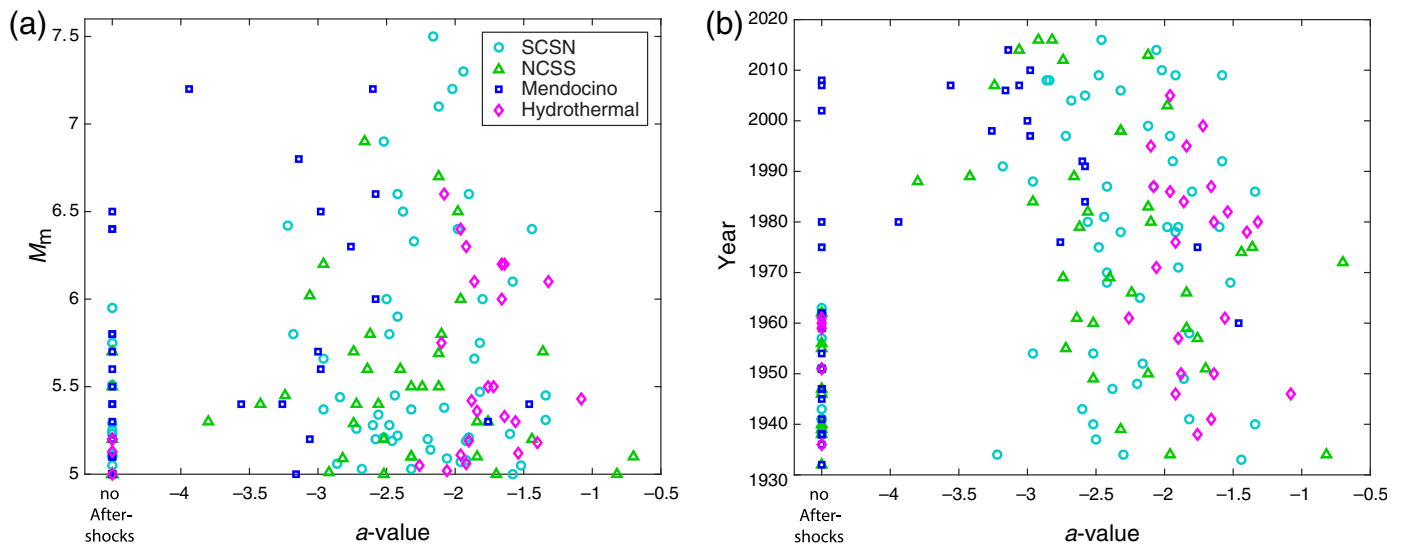
The parameter values found by the joint inversion for mainshocks from 1980 to 2017 in the SCSN region are $a = -2.04$, $p = 0.83$, and $c = 0.0033$ days and for the NCSS region are $a = -2.32$, $p = 0.96$, and $c = 0.012$ days (Fig. 2). We compute the Bayesian information criterion (Schwarz, 1978), $\text{BIC} = \ln(n)k - 2\text{LL}$, in which k is the number of estimated parameters and n is the number of observations. We compare the BIC for the case of separate parameter values for the SCSN and NCSS regions with the BIC for a single set of parameter values for the combined region. The BIC for the separate regions is substantially lower ($-55,706$ compared with $-54,875$, in which a difference of 2 indicates a significant difference at 95% confidence), implying that the use of different a -, p -, and c -values for the two regions is justified by the improvement of the LL. This confirms the result of Llenos and Michael (2017) that southern California sequences are more productive on average than northern California sequences. The Mendocino area is much less productive with $a = -2.80$, $p = 1.15$, and $c = 0.05$ days. The hydrothermal areas in Long Valley, Coso, and the Salton Sea, in contrast,



▲ **Figure 2.** The stacked rate of aftershocks following $M \geq 5$ mainshocks from 1980 to 2017, as a function of time t after the mainshock. Each aftershock is included in the stack at its time after its respective mainshock. Each point shows the aftershock rate found from two consecutive aftershocks in the stack, i and $i + 1$, with rate defined as $1/(t_{i+1} - t_i)$. The rate of aftershocks with magnitude $M \geq M_{\text{cat}}$ and the rate of aftershocks with $M \geq M_c(t, M_m, M_{\text{cat}})$ are shown, in which M_m is the mainshock magnitude, M_{cat} is the background catalog magnitude of completeness, and $M_c(t, M_m, M_{\text{cat}})$ is the time-dependent magnitude of completeness given in equation (2). The curve shows the aftershock rate from the Reasenberg and Jones (1989) model with the parameters that jointly maximize the log likelihood for all sequences, shown using the equivalent magnitude for the stack $M_{\text{equiv}} = (1/b) \log_{10} \sum_{j=1}^K 10^{(bM_{mj})}$, in which j sums over the K mainshocks in the stack. (a) The SCSN region excluding the Salton Sea, Coso, and the El Mayor aftershock sequence; (b) the NCSS region excluding Mendocino and Long Valley; (c) the Mendocino region; and (d) the three hydrothermal regions stacked together: Long Valley, Coso, and the Salton Sea. The color version of this figure is available only in the electronic edition.

are much more productive, with $a = -1.86$, $p = 0.94$, and $c = 0.026$ days. Reasenberg and Jones (1989) noted a higher a -value in eastern California, which may correspond to these hydrothermal areas. The joint inversions for mainshocks from 1932 to 1979 give similar a -values to those found for 1980–2017: $a = -2.04$ for the SCSN region, $a = -2.10$ for the NCSS region, $a = -2.56$ for the Mendocino area, and $a = -1.74$ for the hydrothermal areas.

For comparison, Page *et al.* (2016) find $a = -2.16$, $p = 0.98$, and $c = 0.018$ days for the joint inversion for the active nonsubduction region - shallow continental (ANSR-SHALCON) global tectonic region, which includes most of California (excluding Mendocino). These parameter values are similar to those found for the SCSN and NCSS regions, with the a -value close to the average of the two. For another comparison, we used the California epidemic type aftershock



▲ **Figure 3.** The [Reasenberg and Jones \(1989\)](#) a -values found for individual aftershock sequences of $M \geq 5$ mainshocks from 1933 to 2017. Symbols indicate the different regions in Figure 1. Mainshocks with no aftershocks with $M \geq M_c(t, M_m, M_{cat})$ during the first 10 days, for which the a -value cannot be found, are plotted at the left of each panel. (a) The a -value versus the mainshock magnitude M_m . (b) The a -value versus the mainshock year. The color version of this figure is available only in the electronic edition.

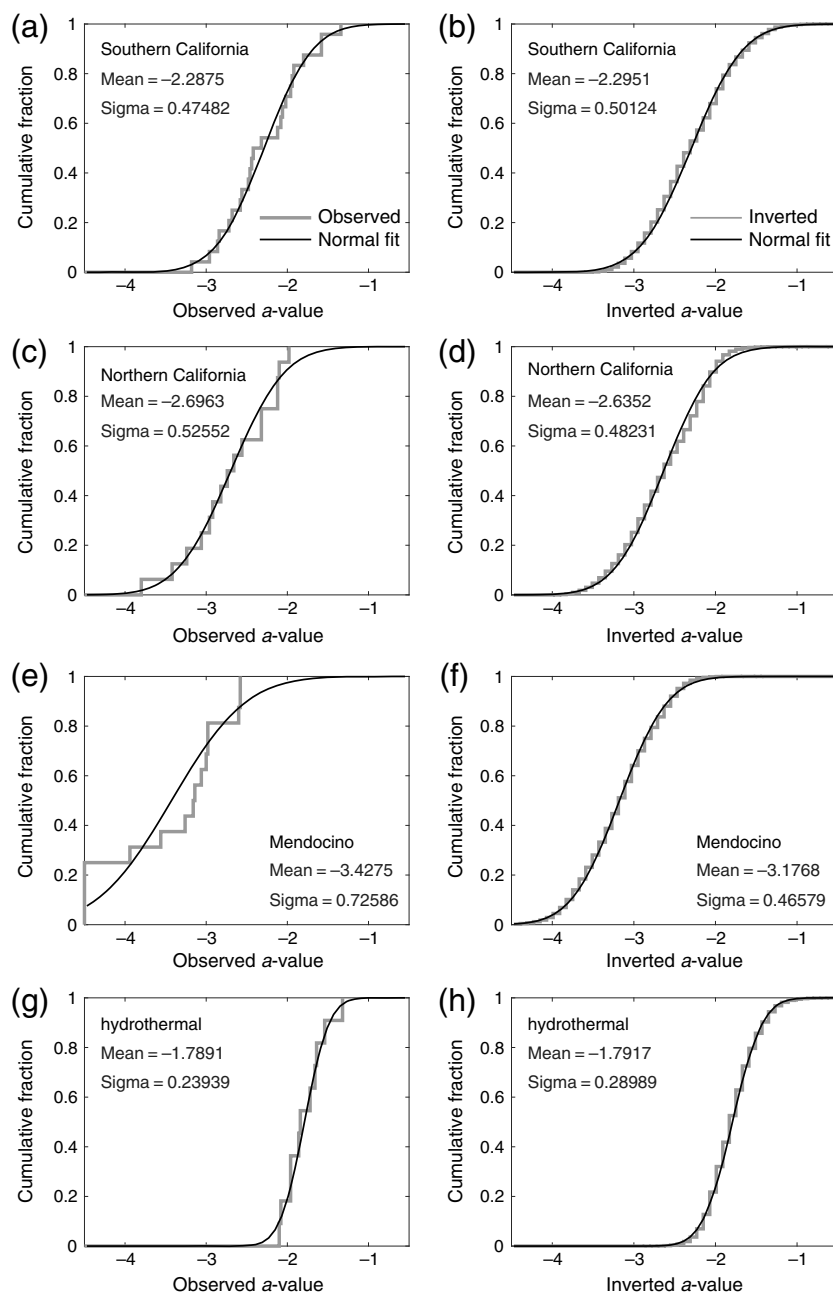
sequence (ETAS) parameters of [Hardebeck \(2013\)](#) to generate synthetic ETAS aftershock sequences. We generate 1000 synthetic sequences following an M_m 7 mainshock and then fit the [Reasenberg and Jones \(1989\)](#) parameters to the first 10 days of the sequences. We find a mode of $a = -2.46$ and $p = 1.03$ (with $b = 0.99$ and $c = 0.0065$ days fixed), again quite close to the parameter values for the SCSN and NCSS regions and probably less productive because the joint inversions are dominated by the highest productivity sequences.

The a -values of the individual sequences are shown in map view (Fig. 1), as a function of mainshock magnitude (Fig. 3a) and as a function of mainshock year (Fig. 3b). There does not appear to be any trend of mean a -value with magnitude. Because there are so few larger magnitude mainshocks, it is difficult to determine whether the a -value variability decreases with increasing magnitude, like [Page et al. \(2016\)](#) observed for the global catalog. The a -values for the $M > 7$ events appear more tightly clustered, but this is likely an effect of the very small number of mainshocks not sampling the tails of the a -value distribution. The apparent differences in a -value with time are largely an artifact of the higher magnitude of com-

pleteness for earlier mainshocks, making it impossible to resolve lower a -values. Changes in network operations and in how magnitude is estimated could also be a source of changes in a -value with time. There are no apparent sudden shifts, nor any obvious drift of a -value with time, for either the SCSN or NCSS regions. The CISCN adopted more consistent measures of magnitude in 2011 ([Uhrhammer et al., 2011](#)), and although there are few mainshocks since then, the difference in a -value between the SCSN and NCSS regions appears to persist after this change.

The distributions of the individual sequence a -values for 1980–2017 are somewhat sparse, and the inversion considerably smoothes them (Fig. 4). In general, the means and standard deviations of the a -value distributions are very similar for the observed and inverted a -values. The inversion makes a substantial difference for Mendocino, which is the only region with many mainshocks that have no aftershocks with magnitude $\geq M_c(t, M_m, M_{cat})$. The mean a -values of the distributions are lower than the a -values found from jointly inverting all sequences because the joint inversions are dominated by the higher productivity aftershock sequences. The inverted a -value distributions are not perfectly normally

Table 2 Generic Parameter Values for Regions				
Region	a , mean	a , 1-sigma	p	c (days)
SCSN	−2.30	0.50	0.83	0.0033
NCSS	−2.64	0.48	0.96	0.012
Mendocino	−3.18	0.47	1.15	0.050
Hydrothermal (Long Valley, Coso, and Salton Sea)	−1.79	0.29	0.94	0.026
SCSN, Southern California Seismic Network; NCSS, Northern California Seismic System.				



▲ **Figure 4.** The cumulative distribution of the a -values for individual aftershock sequences of $M \geq 5$ mainshocks from 1980 to 2017 for each region. The panels on the left show the observed a -values, and the panels on the right show the estimated underlying distribution of true a -values found from inversion. The distributions of the inverted a -values are step-like because of the discretization of the inversion. A normal distribution fit to the a -value distribution is shown. (a,b) The SCSN region excluding the Salton Sea and Coso; (c,d) the NCSS region excluding Mendocino and Long Valley; (e,f) the Mendocino region; and (g,h) the three hydrothermal regions combined: Long Valley, Coso, and the Salton Sea.

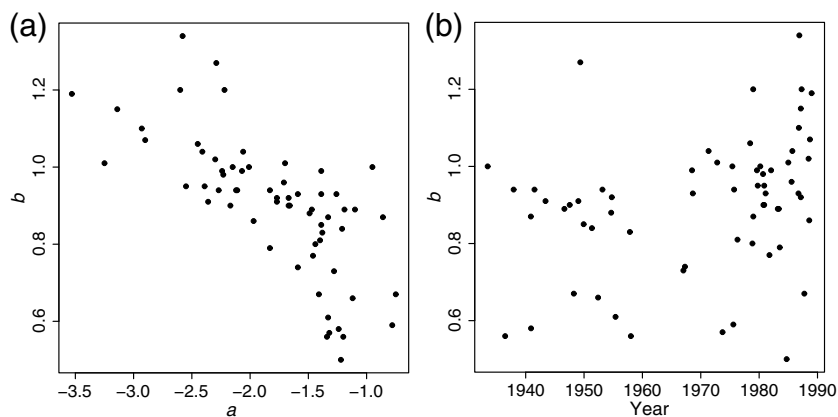
distributed. The small deviations from a normal distribution are unlikely to make much difference to the forecasts, so we choose to approximate all of the inverted distributions with a normal distribution. The preferred generic parameter values for each region are given in Table 2.

With the exception of the hydrothermal regions, our generic a -values are substantially lower than the Reasenberg and Jones (1989) generic a -value. The difference may be because Reasenberg and Jones (1989) omitted mainshocks with too few aftershocks to constrain the b -value and p -value, thereby omitting sequences with low a -values. We test this idea using our catalog for the same time period that Reasenberg and Jones (1989) studied, 1933–1987, to match their catalog completeness. We find that the median a -value for all events in this time period is -2.50 , but the median a -value for all events with at least one aftershock is -1.98 . Reasenberg and Jones (1989) presumably required more than one aftershock to fit the parameters. If we require at least three aftershocks, 60 sequences remain, similar to the 62 sequences used by Reasenberg and Jones (1989), although not the identical set of sequences because of differences in mainshock and aftershock selection. The median a -value of these sequences is -1.91 . This suggests that an ~ 0.6 difference in a -value can be attributed to omitting sequences with few aftershocks. This is very similar to the difference between the Reasenberg and Jones (1989) generic a -value (-1.67) and our generic a -value for southern California (-2.30).

The generic a -value of Reasenberg and Jones (1989) may also have been biased because of a strong trade-off between a -value and b -value, which can be seen in their data set (L. M. Jones, written comm., 2009) (Fig. 5a). They computed individual b -values for each sequence based on all aftershocks with $M \geq 2.0$. This could pose a problem for older sequences, in which the magnitude of completeness is substantially greater than 2.0, as well as for larger mainshocks with substantial short-term incompleteness. There is a paucity of b -values ≥ 1.0 before ~ 1970 , consistent with a bias caused by incompleteness at lower magnitudes (Fig. 5b). This suggests that before ~ 1970 , catalog incompleteness did lead to an underestimate of b -value, and because of the trade-off, a corresponding overestimate of a -value.

The p -values are all close to the typical value of 1.0. Most are < 1.0 , which means that sequences theoretically fail to converge as time goes to infinity, although in practice, the aftershock rates are reasonable over the finite time periods for

which aftershock forecasts are planned. The c -values range from ~ 5 min for the SCSN region to more than an hour for Mendocino. The c -values likely reflect incompleteness in the earliest part of the aftershock sequence because true c -values are likely ≤ 1 min (Enescu *et al.*, 2009). Incompleteness is also



▲ **Figure 5.** The individual-sequence a - and b -values from [Reasenberg and Jones \(1989\)](#) (L. M. Jones, written comm., 2009). (a) The trade-off between a -value and b -value. (b) The b -value as a function of mainshock year.

suggested by the larger c -value in the more sparsely instrumented Mendocino region.

We compare the rate of $M \geq (M_m - 3.0)$ aftershocks for the new generic parameters and their cumulative number with that of the original [Reasenberg and Jones \(1989\)](#) parameters, the modified [Reasenberg and Jones \(1989\)](#) parameters with $a = -1.85$, the ANSR-SHALCON parameters from [Page et al. \(2016\)](#), and the fit to the ETAS simulations. Aside from the original [Reasenberg and Jones \(1989\)](#) parameters, all fall close to the SCSN parameters in both rate and cumulative number (Fig. 6), with the ETAS simulation being the most similar. The ANSR-SHALCON rate is approximately the average of the SCSN and NCSS rates. We also compare the SCSN and NCSS rates with the rates from the individual sequence parameters found by [Llenos and Michael \(2017\)](#) for southern and northern California events, respectively. The individual event rates mostly fall within the 1-sigma bounds of the SCSN or NCSS rates, confirming that we have adequately captured the sequence variability.

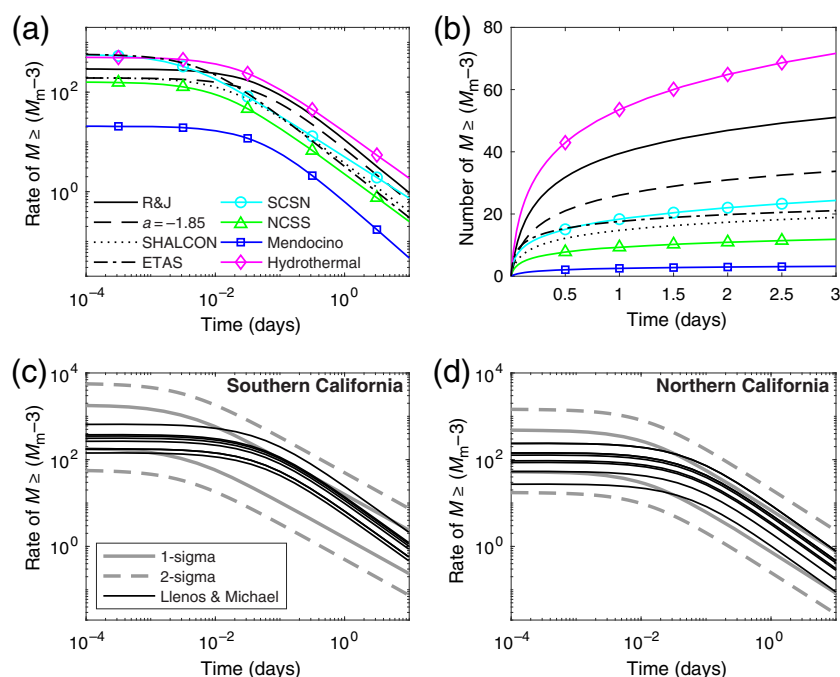
DISCUSSION

Our results imply substantially lower aftershock productivity than the [Reasenberg and Jones \(1989\)](#) generic a -value of -1.67 . The proposed corrected value of -1.85 (e.g., [Michael, 2012](#)) appears more reasonable for southern California but is still high for northern California. The difference in productivity is likely mostly due to [Reasenberg and Jones \(1989\)](#) excluding the lower a -value sequences and possibly the trade-off between a -value and b -value in their parameter fits. There may also be differences in the productivity due to differences in how the spatial extent of the aftershock sequence is defined. Here, we use a simple circular area around the mainshock, with the radius scaling with the [Wells and Coppersmith \(1994\)](#) rupture length of the mainshock, but [Reasenberg and Jones \(1989\)](#) use the [Reasenberg \(1985\)](#) declustering algorithm to identify the aftershocks.

The lower aftershock productivity for the updated parameters raises the issue of whether foreshock and aftershock probabilities are consistent. [Jones \(1985\)](#) found that 6% of $M \geq 3$ non-aftershock earthquakes in southern California, occurring from 1932 to 1983, were foreshocks to a larger event within 10 km and the next five days. This agrees well with the [Reasenberg and Jones \(1989\)](#) parameters with $a = -1.85$, which imply a 7% chance of an earthquake having an aftershock of the same size or larger than itself within five days. This similarity supports the hypothesis that foreshock–mainshock triggering is the same process as mainshock–aftershock triggering ([Jones et al., 1995](#); [Felzer et al., 2004](#)). The updated SCSN parameters, in contrast, give only a 3% chance of an event triggering an aftershock of the same or larger magnitude in the next five days. The foreshock definition in [Jones \(1985\)](#) is different from how aftershocks were defined when developing the SCSN parameters, however, particularly in their use of the 10-km radius for the foreshocks, as opposed to the radius based on the [Wells and Coppersmith \(1994\)](#) rupture length that we use. The earthquakes studied by [Jones \(1985\)](#) also include the Coso and Salton Sea areas, which have higher productivity. We consider $M \geq 3$ nonaftershock earthquakes within the SCSN region, with Coso and the Salton Sea areas removed. We find that 4% of these events are followed by an aftershock of the same size or larger within five days and three times the [Wells and Coppersmith \(1994\)](#) distance, similar to the 3% implied by the SCSN parameters. Therefore, the foreshock and aftershock probabilities are in reasonable agreement if they are defined consistently. However, because three times the [Wells and Coppersmith \(1994\)](#) distance is < 1 km for an M 3.0 earthquake, this is not a very practical definition for the purposes of foreshock statistics.

The relatively high aftershock productivity for the hydrothermal areas is in contrast with the results of [Yang and Ben-Zion \(2009\)](#), who find an inverse relationship between heat flow and aftershock productivity for selected locations in southern California. Some of the $M \geq 5$ mainshocks occur during swarms, so it is possible that what appears as high productivity might actually be high background rate. However, if many of the apparent aftershocks were background events, we would expect to see a less pronounced temporal decay, expressed as a low p -value, which is not observed. Interestingly, the Geysers geothermal field in northern California may not share the high productivity. There is only one mainshock at the Geysers, an M_m 5.0 event in December 2016. The observed a -value for that event is -2.92 , and there are only four $M \geq 2.3$ aftershocks during the first 10 days after the mainshock.

There are other apparent spatial variations in a -value within the SCSN and NCSS regions that are currently unexplained, which may relate to mainshock stress drop and/or the local physical conditions. For global subduction zones, [Wetzler et al. \(2016\)](#) found an inverse relationship between stress drop and aftershock productivity. However, there is no clear



▲ **Figure 6.** The forecast rate or cumulative number of $M \geq (M_m - 3)$ aftershocks from the [Reasenberg and Jones \(1989\)](#) model, with the updated generic parameters for each region. For comparison, the forecast rate or cumulative number from other studies is also shown. (a) The forecast aftershock rates for each of the regions along with the original [Reasenberg and Jones \(1989\)](#) parameters, the modified [Reasenberg and Jones \(1989\)](#) parameters with $a = -1.85$, the active nonsubduction region - shallow continental (ANSR-SHALCON) parameters from [Page et al. \(2016\)](#), and the fit to epidemic type aftershock sequence (ETAS) simulations using the [Hardebeck \(2013\)](#) ETAS parameters. (b) The forecast cumulative number of aftershocks for each of the regions along with the original [Reasenberg and Jones \(1989\)](#) parameters, the modified [Reasenberg and Jones \(1989\)](#) parameters, ANSR-SHALCON, and the fit to the ETAS simulations. (c) The forecast aftershock rate for the SCSN region along with the individual sequences in southern California from [Llenos and Michael \(2017\)](#). The forecast for the SCSN region is shown using the 1-sigma and 2-sigma a -values of the normal distribution from [Figure 4b](#). (d) The forecast aftershock rate for the NCSS region along with the individual sequences in northern California from [Llenos and Michael \(2017\)](#). The forecast for the NCSS region is shown using the 1-sigma and 2-sigma a -values of the normal distribution from [Figure 4d](#). The color version of this figure is available only in the electronic edition.

correlation between the spatial distribution of a -values ([Fig. 1](#)) and the spatial stress-drop variations in southern California of [Shearer et al. \(2006\)](#). For the purposes of aftershock forecasting, the unexplained variations of the a -values within regions are currently best treated as epistemic uncertainty.

CONCLUSION

We update generic parameter values for the [Reasenberg and Jones \(1989\)](#) model of aftershock rate following a mainshock in California. We find productivity values for much of the SCSN and NCSS authoritative regions that are lower than the generic productivity reported by [Reasenberg and Jones](#)

(1989), likely because lower productivity sequences were omitted from their analysis and possibly because of the trade-off of productivity with b -value. The forecast rate of aftershocks from our parameters is similar to the rates inferred from prior studies ([Page et al., 2016](#); [Llenos and Michael, 2017](#)). The Mendocino region exhibits substantially lower aftershock productivity than the statewide average, and the combined hydrothermal areas in Long Valley, Coso, and the Salton Sea exhibit substantially higher aftershock productivity.

The new generic parameters should help improve the accuracy of the aftershock forecasts being issued by the USGS and its partners in the CISEN. The good agreement between the parameter values for the SCSN and NCSS regions and the global ANSR-SHALCON parameters validates the usefulness of the [Page et al. \(2016\)](#) approach of using the global tectonic regionalization to determine generic parameters. This gives us more confidence in using the [Page et al. \(2016\)](#) generic parameters in regions of the United States where local parameter values are not available. However, the large parameter differences for some regions, such as the hydrothermal areas, imply that finer scale regionalization for places of interest can be important.

DATA AND RESOURCES

The earthquake catalog for southern California was accessed from the Southern California Earthquake Data Center (SCEDC, http://service.scedc.caltech.edu/eq-catalogs/date_mag_loc.php, last accessed August 2017). The earthquake catalog for northern California was accessed through the Northern California Earthquake Data Center (NCEDC), doi: 10.7932/NCEDC (<http://www.ncedc.org/ncedc/catalog-search.html>, last accessed August 2017). The California Integrated Seismic Network (CISEN) authoritative regions were accessed through the NCEDC (<http://www.ncedc.org/anss/anss-detail.html#regions>, last accessed July 2017). ☒

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